MST IS222 S12024 Possible Solutions

Web Applications Development (The University of the South Pacific)

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Possible Solutions

Student Name: _____

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Mode: _____

Mid-Semester Test

IS222 – Database Management Systems

Semester I, 2024

Mode: Face-to-Face & Online Students

Date: Thursday, 25th April 2024

Time: 3.00 –3.55 PM (FJ Time)

Weight: 25%

You have 5 minutes to read this paper,

Followed by 50 minutes to complete this paper.

All Questions Are Compulsory

This Test 1 will focus on the following learning outcomes for this course:

CLO1. Explain the fundamental concepts of various database management systems. CLO3. Construct a data model for a relational database management system.

Time allowed: 55 minutes only

Total Marks: 50

Total Pages: 7

Weight: 25%

INSTRUCTIONS:

1. This exam is worth 25% of your coursework.
2. ALL questions are compulsory.
3. Write your Name & ID # on all sheets in this booklet.
4. Zero for any Plagiarism case.

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SECTION A – [10 Marks]

1 Explain what do we mean by "business rules" within the context of a database, and how do they impact database operations? [2 marks]

Business rule: (1 mark)

A business rule is an unambiguous description of business processes.

Or

A business rule is a statement that imposes some form of constraint on a specific aspect of the database, such as the elements within a field specification for a particular field or the characteristics of a given relationship.

Or

A business rule is a brief, precise, and unambiguous description of a policy, procedure, or principle.

Impact business operations: (1 mark)

By incorporating business rules: data remains accurate, reliable, and aligned with their business processes. This integration facilitates efficient data management and decision-making while maintaining compliance with organizational policies and external regulations.

or

- Help standardize company's view of data
- Communications tool between users and designers
- Allow designer to:
  - Understand the nature, role, scope of data, and business processes
  - Develop appropriate relationship participation rules and constraints
  - Create an accurate data model

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2 What strategies can the database designer use to identify the business rules? [2 marks]

Any two should be fine (1 mark each)

- Interviews with Stakeholders

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- Review Documentation and Policies
- Data Profiling and Analysis
- Observation and Workflow Analysis
- Consultation with Domain Experts
- Prototyping and Feedback Sessions

3 Consider the following table:

Project Number	Project Name	Employee Number	Employee Name	Department	Hours Spent
1	Apollo	1001	Smith	Engineering	127
1	Apollo	1003	Jones	Accounts	45
2	Mercury	1002	Stewart	Marketing	70
3	Venus	1001	Smith	Engineering	21
3	Venus	1006	Brown	Engineering	124

Identify the potential problems (at least 2) that may arise if the data was kept in this relation.

[Hint Identifying anomalies in relations] [1mark]

- Data redundancy (duplication) leading to insert, update and delete anomalies.
- Usability Issues - users may have difficulty accessing the correct version of the data or may be confused by inconsistencies and errors. etc.

4 Normalization is a process within logical design. Explain the main objective normalization process. [1 mark]

The main objective is to remove redundancy in data.

5 Ensuring data integrity is an important issue in the database design and it should be addressed at each and every step of the design process. Explain two features of the database design that embodies data integrity concerns. [4 marks]

Minimum cardinality constraints (identified during conceptual design) indicate the minimum number of relationships that MUST exist.

- During physical design, the definition of each attribute can include constraints on the values of the attribute.

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- During logical design, the specification of referential integrity constraints indicates the required relationships among foreign keys and primary keys.

2 marks for each

SECTION B – [20 Marks]

6 Tracking production is crucial across various manufacturing sectors such as pharmaceuticals and children's toys. The ER diagram below depicts key relationships involved in production tracking, including those between production batches, individual units, and raw materials.

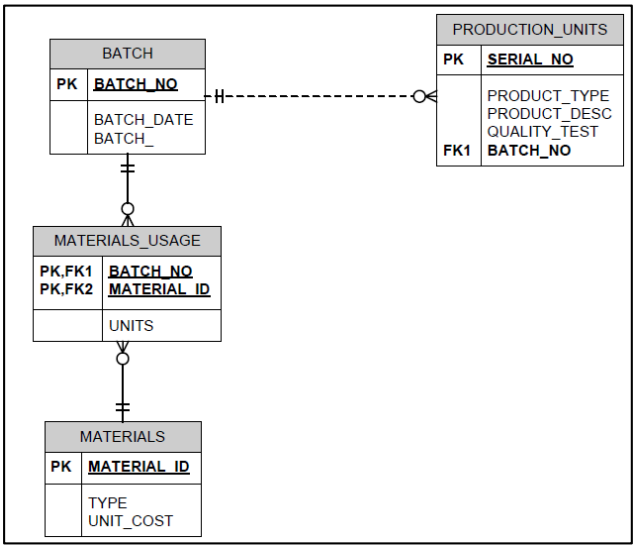

Figure 1: ER diagram showing key relationships involved in production tracking.

a) Identify an attribute in the above ER diagram that might represent a composite attribute, and explain why/how it might represent a composite attribute. [1 mark]

Many of the attributes could actually represent composite attributes:

- Product Description might have several components

- Product type might have several components

- BATCH_DATE might be stored as both date and time.

Mark any one attribute

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b) Now, identify an attribute in the ER diagram that could represent a derived attribute and explain why/how it might represent a derived attribute. [2 marks]

BATCH_TOTALCOST (associated with the Batch entity) most likely represents a derived attribute. The cost of materials could be computed based on the materials unit cost (from the Materials entity) and the number of units required for a batch (on the relationship).

c) The ER diagram/relational database schema contains several instances of data redundancy. Identify and explain one instance where a data redundancy issue exists. [2 marks]

Production type and production description are being stored on each and every production unit, when these parameters are most likely the same for the entire batch.

7 XYZ is a courier company in Fiji. XYZ prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, XYZ relies on a company-wide information system. Shipped items are the heart of the XYZ product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date. Shipped items are received into the XYZ system at a single retail center. Retail centers are characterized by their type, uniqueID, and address. Shipped items make their way to their destination via one or more standard XYZ transportation events (i.e., flights, truck deliveries). These transportation events are characterized by a unique scheduleNumber, a type (e.g, flight, truck), and a deliveryRoute.

a) Write the possible business rules. [6 marks]

(3 marks each)

1. An item can be shipped via one or more transportation event. A transportation event can ship one or more items.
2. A retail center receives one (zero) or many items. An item is received by only one retail center.

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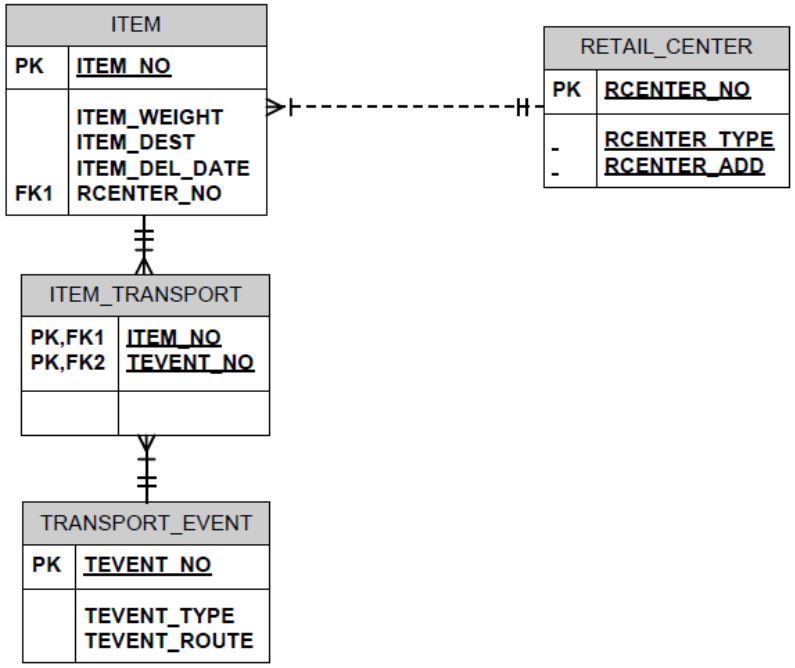
b) Create an Entity Relationship diagram that captures this information about the XYZ Company. [9 marks]



- 1.5 Marks for each entity – PK/FK, Attribute, Entity name (0.5 mark each)
- 2 Marks for relationship – cardinality, connectivity, strength

SECTION C – [20 Marks]

Table 1: The medical history record.

PET ID	PET NAME	PET TYPE	PET AGE	OWNER	VISIT DATE	PROCEDURE
246	ROVER	DOG	12	SAM COOK	• Jan 12/2002 • Mar 27/2002 • Apr 02/2002	• 01-Rabies Vaccination • 10 – Examine and Treat Wound • 05- Heart Worm Test
298	SPOT	DOG	2	TERRY KIM	• Jan 21/2002 • Mar 10/2002	• 08- Tetanus Vaccination • 05- Heart Worm Test

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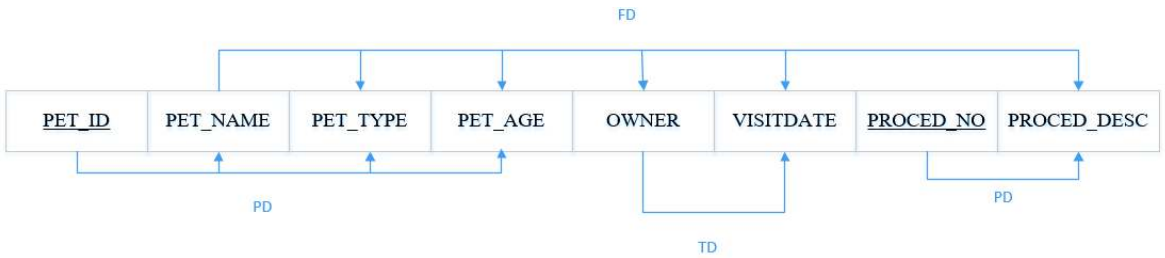
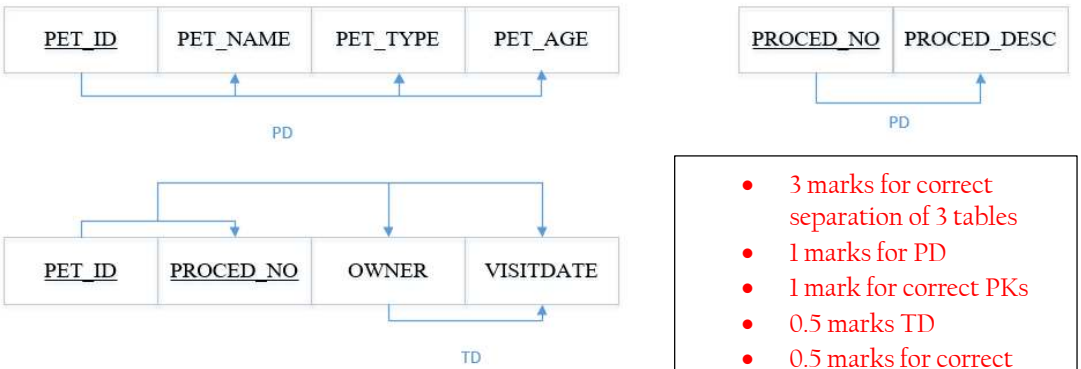
341	MORRIS	CAT	4	SAM COOK	• Jan 12/2002 • Mar 27/2002 • Apr 02/2002	• 01- Rabies Vaccination • 01- Rabies Vaccination
519	TWEEDY	BIRD	2	TERRY KIM	• Jan 12/2002 • Mar 27/2002 • Apr 02/2002	• 20- Annual Check Up • 12- Eye Wash

a) Draw the dependency diagram showing all types of dependencies that is functional, partial, and transitive. [6 marks]



- 1.5 marks PK
- 2 marks for PD
- 0.5 marks TD
- 2 marks FD

b) Using the initial dependency diagram, you have given above, convert your table structure to 2NF, and draw the new dependency diagram. [6 marks]



- 3 marks for correct separation of 3 tables
- 1 marks for PD
- 1 mark for correct PKs
- 0.5 marks TD
- 0.5 marks for correct flow.

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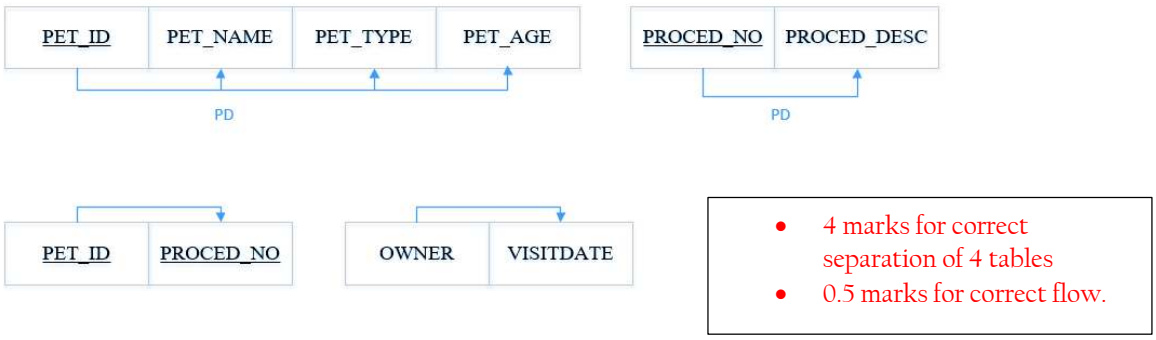
c) Using the table structures you created above, convert your table structure to 3NF, and draw the new dependency diagram. [6 marks]



- 4 marks for correct separation of 4 tables
- 0.5 marks for correct flow.

d) Normalization is a process within logical design. Explain the main objective normalization process. [2 marks]

Normalization is a crucial process of logical database design.
Its primary objective is to reduce data redundancy and dependency, ensuring that data is organized efficiently and accurately.
By breaking down larger tables into smaller, more manageable ones and establishing relationships between them through keys, normalization aims to eliminate anomalies such as insertion, update, and deletion anomalies.
Ultimately, normalization enhances data integrity, making it easier to maintain and query databases while minimizing the risk of inconsistencies.

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All the Best!
The End …

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